Engine Model(s): PT6A

70-00-00

STANDARD PRACTICES - CLEANING

1. General

A. Gas turbine cleaning can be categorized into two broad areas, namely specific parts cleaning and field cleaning (i.e., compressor washing).

B. Specific parts cleaning is primarily to remove contaminants which might conceal minor cracks and other defects which, if not detected, could eventually lead to failure of a component or part. Engine components or parts should be cleaned only as necessary to perform required inspection and repair. Overcleaning of components or parts is unnecessary and should be avoided. The cleaning methods given in the following text are adequate for all maintenance levels. For compressor washing methods, refer to Chapter 71-00-00, CLEANING.

2. Cleaning

A. General

(1) Prior to engine disassembly, ensure that the work area is clean.

(2) Benches on which engine parts are placed must be clean and free from grit, metal filings, etc., which may contaminate engine oil systems, fuel systems or hardware.

(3) Clean plastic bags should be available in which oil system and fuel system parts may be stored until ready for reassembly.

(4) Clean plastic caps or covers should be used to protect exposed tubes or bearing areas.

(5) At reassembly make sure that all parts are clean and new packings installed.

B. Precautions

NOTE: The choice of cleaning agents should be limited to the consumables listed in CONSUMABLE MATERIALS section. The toxicity of cleansing agent will, however, depend on the type of contamination encountered on part to be cleaned.

CAUTION: TAKE PARTICULAR CARE IN SELECTING CLEANING METHOD TO MAKE SURE THAT ANODIZING AND OTHER PROTECTIVE COATINGS ARE NOT REMOVED FROM PARENT METAL. DO NOT USE ALKALIES ON ALUMINUM, MAGNESIUM, ALUMINIZED AND PAINTED AREAS.

(1) Wear rubber gloves, apron or coveralls and face shield or goggles, when working with solvents.

(2) Use the least toxic of available cleaning materials which will satisfactorily accomplish work.

(3) Do all cleaning operations in a well ventilated work area.

(4) Make sure that adequate and usable fire fighting and safety equipment is conveniently located and available to all personnel.

(5) Do not smoke or expose a flame within 50 feet of cleaning area.

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(6) Make sure that all degreasing agents are thoroughly removed from all parts after cleaning.

(7) Do not use steel brushes for any cleaning operation except when specifically detailed within this manual. Use a stiff bristle fiber brush.

C. Cleaning Before Welding

(1) Surface to be welded must be free from protective coatings, dirt, grease, oil and other contaminants, and as free as practicable from oxide formation. Wire brushes and abrasives may be used to remove protective coatings and oxides, except that the final step in removing oxides from aluminum alloys preferably should consist of chemical treatment immediately prior to welding. Wire brushes, when used for cleaning corrosion-resistant alloy, must have bristles of austenitic, corrosion-resistant steel. No undesirable deposit or residue must remain on surface to be welded after cleaning operation.

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